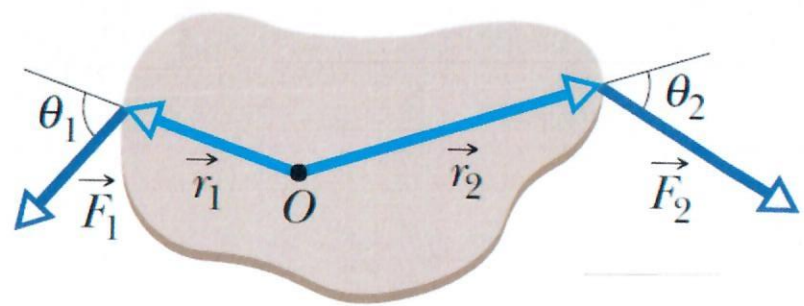
1. A disk rotates about its central axis starting from rest and accelerates with constant angular acceleration. At one time it is rotating at 10 rev/s; 60 revolutions later, its angular speed is 15 rev/s. Calculate
   (a) the angular acceleration,
   (b) the time required to complete the 60 revolutions,
   (c) the time required to reach the 10 rev/s angular speed.

'n Skyf begin uit rus en om roteer sy sentrale as en versnel met 'n konstante hoek versnelling. Op een tydstip roteer die skyf teen 10 rev/s; 60 revolusies later is die hoekspoed 15 rev/s. Bereken
(a) die hoek versnelling,
(b) die tyd wat dit neem om 60 revolusies te voltooi,
(c) die tyd wat dit neem om 'n hoekspoed van 10 rev/s te bereik. (7)

2. The body in the figure is pivoted at O, and two forces act on it as shown. If $r_1 = 1.30$ m, $r_2 = 2.15$ m, $\vec{F}_1 = 4.20$ N, $\vec{F}_2 = 4.90$ N, $\theta_1 = 75.0°$, and $\theta_2 = 60.0°$, what is the net torque about the pivot?



Die voorwerp in die figuur draai om die punt O, en twee kragte werk in op die voorwerp soos aangedui. Indien $r_1 = 1.30$ m, $r_2 = 2.15$ m, $\vec{F}_1 = 4.20$ N, $\vec{F}_2 = 4.90$ N, $\theta_1 = 75.0°$, en $\theta_2 = 60.0°$, wat is die netto wringkrag om die draaipunt? (3)

Tutorial Test 9-1

1. $\omega_1 = 10\ \text{rev/s} = 10\ \text{rev/s} \times 2\pi \frac{\text{rad}}{\text{rev}} = 20\pi\ \text{rad/s}$

$\omega_2 = 15\ \text{rev/s} = 15\ \text{rev/s} \times 2\pi \frac{\text{rad}}{\text{rev}} = 30\pi\ \text{rad/s}$

$\Delta\theta = \theta_2 - \theta_1 = 60\ \text{rev} = 120\pi\ \text{rad}$

a) $\omega_2^2 = \omega_1^2 + 2\alpha\Delta\theta$

$$\alpha = \frac{\omega_2^2 - \omega_1^2}{2\Delta\theta} = \frac{(30\pi\ \text{rad/s})^2 - (20\pi\ \text{rad/s})^2}{2(120\pi\ \text{rad})}$$

$$= \frac{500\pi^2\ \text{rad}^2/\text{s}^2}{240\pi\ \text{rad}} \left.\begin{aligned} &= 2.1\ \pi\ \text{rad/s}^2 \\ &= 6.5\ \text{rad/s}^2 \end{aligned}\right\} \text{either}$$

b) $\omega_2 = \omega_1 + \alpha t$

$$t = \frac{\omega_2 - \omega_1}{\alpha} = \frac{30\pi\ \text{rad/s} - 20\pi\ \text{rad/s}}{6.54\ \text{rad/s}^2}$$

$\Delta t = 4.8\ \text{s}$

c) $\omega_0 = 0$ at $t = 0$.

$\omega_1 = \omega_0 + \alpha t$

$$t = \frac{\omega_1 - \omega_0}{\alpha} = 9.6\text{s}$$

Tut Test 9-1

2. $F_1 = 4.20\text{ N}$ $\quad r_1 = 1.30\text{ m}$ $\quad \theta_1 = 75.0°$

$F_2 = 4.90\text{ N}$ $\quad r_2 = 2.15\text{ m}$ $\quad \theta_2 = 60.0°$

$$\tau_{net} = \tau_1 + \tau_2$$

$$= r_1F_1 \sin\theta_1 - r_2F_2 \sin\theta_2$$

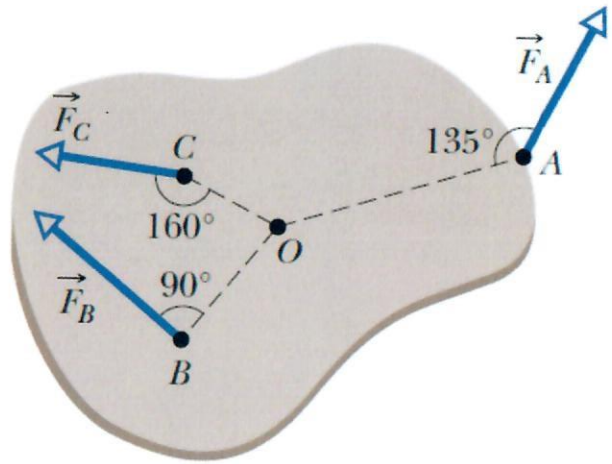
$$= 4.20\text{N}(1.30\text{m})(\sin 75.0°) - 4.90\text{N}(2.15\text{m})(\sin 60.0°)$$

$$= -3.85\text{ N.m}$$

1. The angular position of a point on the rim of a rotating wheel is given by $\theta = 4.0t - 3.0t^2 + t^3$, where θ is in radians and t is in seconds.
   a) What is the angular velocity at $t = 2.0$ s
   b) What is the average angular acceleration for the time interval that begins at $t = 2.0$ s and ends at $t = 4.0$ s?
   c) What is the instantaneous angular acceleration at $t = 2.0$ s?
   d) What is the angular displacement between $t = 2.0$ s and $t = 4.0$ s?

(7)

2. The body in the figure is pivoted at O. Three forces act on it: $F_A = 10$ N at point A, 8.0 m from O; $F_B = 16$ N at B, 4.0 m from O; and $F_C = 19$ N at C, 3.0 m from O. What is the net torque about O?
Die voorwerp in die figuur draai om die punt O, en twee kragte werk in op die voorwerp: $F_A = 10$ N by punt A, 8.0 m van O; $F_B = 16$ N by B, 4.0 m van O; en $F_C = 19$ N by C, 3.0 m van O. Wat is die netto wringkrag om O?



(3)

Tutorial Test 9-2

1. $\theta = 4.0t - 3.0t^2 + t^3$

$\omega = \frac{d\theta}{dt}$

a) $\omega = 4.0 - 6.0t + 3t^2$

At $t = 2.0s$

$\omega = 4.0 - 6.0(2.0s) + 3(2.0s)^2$

$= 4 \text{ rad/s}$

b) $\alpha_{avg} = \frac{\omega_4 - \omega_2}{\Delta t}$

$\omega_4 = 4.0 - 6.0(4.0s) + 3.0(4.0s)^2$

$= 28 \text{ rad/s}$

$\alpha_{avg} = \frac{28 \text{ rad/s} - 4 \text{ rad/s}}{2s} = 12 \text{ rad/s}^2$

c) $\alpha = \frac{d\omega}{dt} = -6.0 + 6.0t$

$\alpha_2 = -6.0 + 6.0(2.0s)$

$= 6.0 \text{ rad/s}^2$

Tut Test 9-2

2. $F_A = 10\ N$; $\phi_A = 45°$; $r_A = 8.0m$

$F_B = 16\ N$; $\phi_B = 90°$; $r_B = 4.0m$

$F_C = 19\ N$; $\phi_C = 30°$; $r_C = 3.0m$

$$\tau_{net} = \tau_A - \tau_B + \tau_C$$
$$= r_A F_A \sin\phi_A - r_B F_B \sin\phi_B + r_C F_C \sin\phi_C$$
$$= 8.0m(10N)\sin 45° - 4.0m(16N)\sin 90° + 3.0m(19N)\sin 20°$$

$$\tau_{net} = (56.6 - 64.0 + 19.5)\ N.m$$
$$= 12.1\ Nm$$

Since $\sin\theta$ is positive for all θ less than 180°
The use of $\sin 135°$ and $\sin 160°$
also give the correct answer